

Now, we prove that $\Psi_c \setminus \{(0, m) : m \geq c\} \subset \Psi_c^*$. For any $0 < p \leq 1$, consider any $(p, m) \in \Psi_c$ and the following probability mass function for a random variable T :

$$\Pr\{T = 0\} = 1 - p \quad \text{and} \quad \Pr\{T = m/p\} = p.$$

Since $m \geq cp$, we have $m/p \geq c$, and so

$$\Pr\{T \geq c\} = \Pr\{T = m/p\} = p,$$

and $E[T] = p(m/p) = m$. Therefore, (p, m) is achieved by the random variable T as constructed. Note that unless $m = cp$, (p, m) can always be achieved by more than one probability distribution.

It remains to prove that every ordered pair $(0, m)$ with $0 \leq m < c$ is achievable. This can be done by noting that such an ordered pair can be achieved by any random variable T with $\Pr\{T = m\} = 1$. The theorem is proved. $\square$

The region Ψ_c is defined by Markov's inequality together with the constraint $0 \leq p \leq 1$ which comes from the setup of the problem, and we have shown that for any fixed $c > 0$, every ordered pair in Ψ_c except for a region with Lebesgue measure 0 (namely the region $\{(0, m) : m \geq c\}$) is achievable by some random variable T . Specifically:

- When $\Pr\{T \geq c\} > 0$, Markov's inequality, namely

$$E[T] \geq c \cdot \Pr\{T \geq c\}, \tag{10}$$

which gives a lower bound on $E[T]$, is the only constraint on $E[T]$ in terms of $\Pr\{T \geq c\}$.

- When $\Pr\{T \geq c\} = 0$, Markov's inequality as in (10), which becomes $E[T] \geq 0$, continues to be valid. However, we also have

$$E[T] < c. \tag{11}$$

Combining (10) and (11), we have

$$0 \leq E[T] < c.$$